1:execCode(apiServer,jetty):0

50:RULE 5 (multi-hop access):0 57:RULE 5 (multi-hop access):0 81:RULE 5 (multi-hop access):0 146:RULE 5 (multi-hop access):0

2:RULE 2 (remote exploit of a server program):0

3:netAccess(apiServer,tcp,8443):0

33:RULE 2 (remote exploit of a server program):0 35:RULE 2 (remote exploit of a server program):0 37:RULE 2 (remote exploit of a server program):0 31:RULE 2 (remote exploit of a server program):0 45:RULE 2 (remote exploit of a server program):0 43:RULE 2 (remote exploit of a server program):0 39:RULE 2 (remote exploit of a server program):0 41:RULE 2 (remote exploit of a server program):0

4:RULE 5 (multi-hop access):0

5:hacl(portalServer,apiServer,tcp,8443):1 6:execCode(portalServer,tomcat):0

7:RULE 2 (remote exploit of a server program):0

8:netAccess(portalServer,tcp,8080):0

27:RULE 2 (remote exploit of a server program):0

9:RULE 5 (multi-hop access):0

11:execCode(reverseProxy,haproxy):0 10:hacl(reverseProxy,portalServer,tcp,8080):1

12:RULE 2 (remote exploit of a server program):0

13:netAccess(reverseProxy,tcp,443):0

21:RULE 2 (remote exploit of a server program):0 19:RULE 2 (remote exploit of a server program):0 23:RULE 2 (remote exploit of a server program):0

14:RULE 6 (direct network access):0

15:hacl(internet,reverseProxy,tcp,443):1 16:attackerLocated(internet):1

74:RULE 22 (Browsing a malicious website):0

22:vulExists(reverseProxy,’CVE-2026-33555’,haproxy,remoteExploit,privEscalation):1 20:vulExists(reverseProxy,’CVE-2026-26081’,haproxy,remoteExploit,privEscalation):1 17:networkServiceInfo(reverseProxy,haproxy,tcp,443,haproxy):1 18:vulExists(reverseProxy,’CVE-2026-26080’,haproxy,remoteExploit,privEscalation):1 24:vulExists(reverseProxy,threat_t1_tampering,haproxy,remoteExploit,privEscalation):1

28:vulExists(portalServer,threat_t2_spoofing,tomcat,remoteExploit,privEscalation):1 25:networkServiceInfo(portalServer,tomcat,tcp,8080,tomcat):1 26:vulExists(portalServer,’CVE-2023-24998’,tomcat,remoteExploit,privEscalation):1

34:vulExists(apiServer,’CVE-2026-12611’,jetty,remoteExploit,privEscalation):1 36:vulExists(apiServer,’CVE-2026-19203’,jetty,remoteExploit,privEscalation):1 30:vulExists(apiServer,’CVE-2026-10050’,jetty,remoteExploit,privEscalation):1 38:vulExists(apiServer,’CVE-2026-19204’,jetty,remoteExploit,privEscalation):1 29:networkServiceInfo(apiServer,jetty,tcp,8443,jetty):1 32:vulExists(apiServer,’CVE-2026-10051’,jetty,remoteExploit,privEscalation):1 46:vulExists(apiServer,threat_t3_information_disclosure,jetty,remoteExploit,privEscalation):1 44:vulExists(apiServer,’CVE-2026-8384’,jetty,remoteExploit,privEscalation):1 40:vulExists(apiServer,’CVE-2026-5795’,jetty,remoteExploit,privEscalation):1 42:vulExists(apiServer,’CVE-2026-6790’,jetty,remoteExploit,privEscalation):1

47:execCode(authServer,keycloak):0

48:RULE 2 (remote exploit of a server program):0

49:netAccess(authServer,tcp,9443):0

51:hacl(apiServer,authServer,tcp,9443):1

52:networkServiceInfo(authServer,keycloak,tcp,9443,keycloak):1 53:vulExists(authServer,threat_t4_spoofing,keycloak,remoteExploit,privEscalation):1

54:execCode(cacheServer,memcached):0

55:RULE 2 (remote exploit of a server program):0

56:netAccess(cacheServer,tcp,11211):0

63:RULE 2 (remote exploit of a server program):0 61:RULE 2 (remote exploit of a server program):0

58:hacl(apiServer,cacheServer,tcp,11211):1

60:vulExists(cacheServer,’CVE-2026-47783’,memcached,remoteExploit,privEscalation):1 59:networkServiceInfo(cacheServer,memcached,tcp,11211,memcached):1 64:vulExists(cacheServer,threat_t6_tampering,memcached,remoteExploit,privEscalation):1 62:vulExists(cacheServer,’CVE-2026-47784’,memcached,remoteExploit,privEscalation):1

65:execCode(clinicianWorkstation,user):0

68:RULE 8 (Access a host through executing code on the machine):0 71:RULE 12 (password sniffing):0

66:RULE 0 (When a principal is compromised any machine he has an account on will also be compromised):0

67:canAccessHost(clinicianWorkstation):0

69:hasAccount(clinician_user,clinicianWorkstation,user):1

72:RULE 3 (remote exploit for a client program):0

70:principalCompromised(clinician_user):0

73:accessMaliciousInput(clinicianWorkstation,clinician_user,thunderbird):0

75:hacl(clinicianWorkstation,internet,httpProtocol,httpPort):1 76:inCompetent(clinician_user):1

77:vulExists(clinicianWorkstation,threat_t8_elevation_of_privilege,thunderbird,remoteClient,privEscalation):1

78:execCode(dbServer,postgresql):0

79:RULE 2 (remote exploit of a server program):0

80:netAccess(dbServer,tcp,5432):0

99:RULE 2 (remote exploit of a server program):0 109:RULE 2 (remote exploit of a server program):0 119:RULE 2 (remote exploit of a server program):0 133:RULE 2 (remote exploit of a server program):0 93:RULE 2 (remote exploit of a server program):0 137:RULE 2 (remote exploit of a server program):0 117:RULE 2 (remote exploit of a server program):0 141:RULE 2 (remote exploit of a server program):0 95:RULE 2 (remote exploit of a server program):0 101:RULE 2 (remote exploit of a server program):0 127:RULE 2 (remote exploit of a server program):0 121:RULE 2 (remote exploit of a server program):0 105:RULE 2 (remote exploit of a server program):0 85:RULE 2 (remote exploit of a server program):0 129:RULE 2 (remote exploit of a server program):0 139:RULE 2 (remote exploit of a server program):0 89:RULE 2 (remote exploit of a server program):0 103:RULE 2 (remote exploit of a server program):0 113:RULE 2 (remote exploit of a server program):0 123:RULE 2 (remote exploit of a server program):0 107:RULE 2 (remote exploit of a server program):0 87:RULE 2 (remote exploit of a server program):0 111:RULE 2 (remote exploit of a server program):0 125:RULE 2 (remote exploit of a server program):0 131:RULE 2 (remote exploit of a server program):0 97:RULE 2 (remote exploit of a server program):0 91:RULE 2 (remote exploit of a server program):0 135:RULE 2 (remote exploit of a server program):0 115:RULE 2 (remote exploit of a server program):0

82:hacl(apiServer,dbServer,tcp,5432):1

100:vulExists(dbServer,’CVE-2026-14671’,postgresql,remoteExploit,privEscalation):1 110:vulExists(dbServer,’CVE-2026-14678’,postgresql,remoteExploit,privEscalation):1 120:vulExists(dbServer,’CVE-2026-15742’,postgresql,remoteExploit,privEscalation):1 134:vulExists(dbServer,’CVE-2026-6464’,postgresql,remoteExploit,privEscalation):1 84:vulExists(dbServer,’CVE-2020-13692’,postgresql,remoteExploit,privEscalation):1 94:vulExists(dbServer,’CVE-2026-14668’,postgresql,remoteExploit,privEscalation):1 138:vulExists(dbServer,’CVE-2026-6470’,postgresql,remoteExploit,privEscalation):1 118:vulExists(dbServer,’CVE-2026-15741’,postgresql,remoteExploit,privEscalation):1 142:vulExists(dbServer,threat_t5_information_disclosure,postgresql,remoteExploit,privEscalation):1 96:vulExists(dbServer,’CVE-2026-14669’,postgresql,remoteExploit,privEscalation):1 102:vulExists(dbServer,’CVE-2026-14672’,postgresql,remoteExploit,privEscalation):1 128:vulExists(dbServer,’CVE-2026-18024’,postgresql,remoteExploit,privEscalation):1 122:vulExists(dbServer,’CVE-2026-16238’,postgresql,remoteExploit,privEscalation):1 106:vulExists(dbServer,’CVE-2026-14676’,postgresql,remoteExploit,privEscalation):1 86:vulExists(dbServer,’CVE-2026-14662’,postgresql,remoteExploit,privEscalation):1 83:networkServiceInfo(dbServer,postgresql,tcp,5432,postgresql):1 130:vulExists(dbServer,’CVE-2026-18408’,postgresql,remoteExploit,privEscalation):1 140:vulExists(dbServer,’CVE-2026-6471’,postgresql,remoteExploit,privEscalation):1 90:vulExists(dbServer,’CVE-2026-14664’,postgresql,remoteExploit,privEscalation):1 104:vulExists(dbServer,’CVE-2026-14673’,postgresql,remoteExploit,privEscalation):1 114:vulExists(dbServer,’CVE-2026-14680’,postgresql,remoteExploit,privEscalation):1 124:vulExists(dbServer,’CVE-2026-16239’,postgresql,remoteExploit,privEscalation):1 108:vulExists(dbServer,’CVE-2026-14677’,postgresql,remoteExploit,privEscalation):1 88:vulExists(dbServer,’CVE-2026-14663’,postgresql,remoteExploit,privEscalation):1 112:vulExists(dbServer,’CVE-2026-14679’,postgresql,remoteExploit,privEscalation):1 126:vulExists(dbServer,’CVE-2026-16241’,postgresql,remoteExploit,privEscalation):1 132:vulExists(dbServer,’CVE-2026-19385’,postgresql,remoteExploit,privEscalation):1 98:vulExists(dbServer,’CVE-2026-14670’,postgresql,remoteExploit,privEscalation):1 92:vulExists(dbServer,’CVE-2026-14666’,postgresql,remoteExploit,privEscalation):1 136:vulExists(dbServer,’CVE-2026-6469’,postgresql,remoteExploit,privEscalation):1 116:vulExists(dbServer,’CVE-2026-14681’,postgresql,remoteExploit,privEscalation):1

143:execCode(mqServer,rabbitmq):0

144:RULE 2 (remote exploit of a server program):0

145:netAccess(mqServer,tcp,5672):0

152:RULE 2 (remote exploit of a server program):0 150:RULE 2 (remote exploit of a server program):0

147:hacl(apiServer,mqServer,tcp,5672):1

149:vulExists(mqServer,’CVE-2019-11287’,rabbitmq,remoteExploit,privEscalation):1 148:networkServiceInfo(mqServer,rabbitmq,tcp,5672,rabbitmq):1 153:vulExists(mqServer,threat_t7_tampering,rabbitmq,remoteExploit,privEscalation):1 151:vulExists(mqServer,’CVE-2020-5419’,rabbitmq,remoteExploit,privEscalation):1